

DoctorLink

Autonomous Hybrid Healthcare for Emerging Markets

Somnia Agentathon Submission

SA Healthcare Crisis

- **Severe Access Disparity:** Public clinics face excessive waiting times, while private care remains prohibitively expensive.
- **Gig Economy Friction:** Doctors desire freelance flexibility but struggle with payment delays and administrative bloat.
- **Telehealth Trust Deficit:** Payments lack Escrow protection, and medical records are vulnerable to breaches.
- **Clinical Safety Gaps:** A critical lack of automated, real-time auditing for dangerous drug-to-drug interactions.



The DoctorLink Solution



Flexible Gig Mode

Doctors easily toggle availability and set custom pricing in South African Rand (fiat) or STT/T-800 utility tokens.



Secure On-Chain Escrow

Patient funds are locked via smart contracts and securely released only upon successful completion of the consultation.



On-Chain AI Agents

Five specialized agents operate as autonomous background tasks, interacting directly with the Somnia L1 blockchain.

System Architecture

Hybrid Orchestration

Our FastAPI backend routes data seamlessly between the patient interface, Supabase relational storage, and decentralized Filebase buckets.

It integrates directly with the Somnia Agentic L1 via custom Solidity Escrow and Sponsor contracts to execute trustless Web3 transactions effortlessly.



On-Chain Agents

Agent Name	Function & Blockchain Interaction
1. Appointment Matcher	Matches waiting patients to doctors in "Gig Mode" based on dynamic pricing and real-time availability.
2. Prescription Reviewer	Calls Somnia LLM (Agent ID 12847...) on-chain to instantly flag potential drug interactions.
3. Follow-Up Scheduler	Invokes Somnia LLM on-chain to generate highly personalized patient post-care instructions.
4. Escrow Agent	Executes on-chain fund releases or refunds natively on the DoctorLinkEscrow contract.
5. AI Doctor Agent	Simulates consultations, uses Somnia LLM for observation, and triggers automated on-chain escrow resolution.

Smart Contract Ecosystem

DoctorLinkEscrow.sol

On-Chain Escrow: Securely locks patient consultation fees prior to appointments.

Platform Fee Split: Automatically splits final payments—80% disbursed to the Doctor, 20% retained as Platform fee.

Safety Lock: Only authorized platform addresses hold the capability to release or refund funds.

DoctorLinkSponsor.sol

STT Pool Sponsor: Securely holds STT specifically to pay for agent gas fees and processing deposits.

Operator Model: Eliminates the security risk of distributing developer private keys. The backend interacts via the sponsor contract.

AgentPaymentRouter: Charges T-800 tokens for advanced execution requirements.

AI Inference Mechanics

DoctorLink leverages native on-chain LLM execution. Background threads systematically poll the validator consensus layer for sub-second, decentralized medical processing without relying on fragile or restrictive centralized API keys.

Payments & Storage



Fiat Payments

Integrates popular South African gateways (PayFast, Yoco). Users pay easily with Credit Card or Instant EFT.

 Crypto Web3 smart contract



Web3 Payments

Frictionless payments via MetaMask or Somnia wallets utilizing STT or native T-800 utility tokens.

 Secure cloud data storage



Filebase Storage

Immutable history of patient medical records and prescriptions stored via decentralized, S3-compatible Filebase.

Why Somnia L1?



High Throughput & Speed

Sub-second block times allow instant patient-doctor matching, real-time messaging, and rapid escrow state updates.



Extremely Low Gas Fees

Deploying contracts, executing splits, and orchestrating complex smart contract logic costs fractions of a cent.



Native On-Chain AI

Directly queries LLMs on the consensus layer. No centralized API keys to leak, protecting strict POPIA privacy regulations.



Decentralized Security

Medical data hashes, financial escrows, and payments are fully audited, transparent, and secured irrevocably on-chain.

Roadmap & Vision

Multilingual AI

Fine-tune Somnia AI Doctors on local languages (isiZulu, isiXhosa, Afrikaans) using custom prompts.

Expansion

Replicate the scalable gig-economy telemedicine model across other emerging markets in Sub-Saharan Africa.

POPIA Compliance

Implement advanced Zero-Knowledge Proofs (ZKP) to verify identity without revealing health metadata.

IoT Integration

Enable agents to listen to real-time wearable telemetry (heart rate) to automatically trigger clinical warnings.

Image Sources



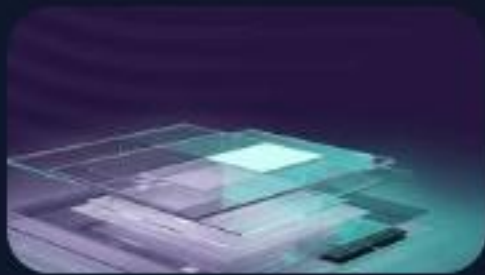
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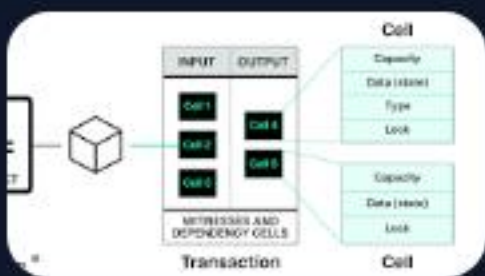
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